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Mid theory (Lecture 05)

In species distribution modeling using MaxEnt (Maximum Entropy), the three main types of projections are usually

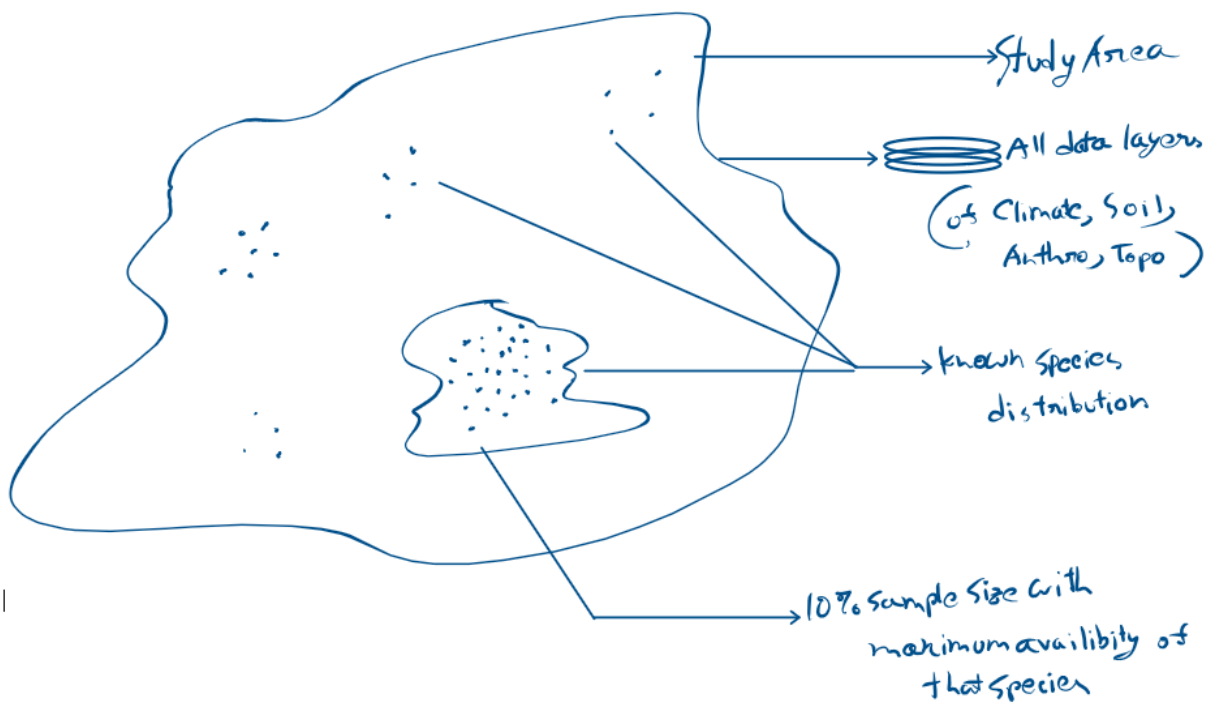
- Current distribution
- Potential distribution
- Projected distribution

The MaxEnt (Maximum Entropy) model requires multiple types of spatial and environmental data to accurately predict species distribution. The first step involves defining the study area, followed by the collection and preparation of relevant datasets for the entire region. Each dataset must be processed into a spatial layer (raster format) to serve as input for the model.

- The parameters typically considered in MaxEnt modeling include the following:
- Climatic factors: temperature, rainfall, humidity, and evapotranspiration
- Soil nutrients: nitrogen (N), phosphorus (P), potassium (K), sulfur (S), pH, zinc (Zn), magnesium (Mg), organic matter, and carbon
- Topographic factor: elevation
- Anthropogenic factors: distance from roads, infrastructure, industrial zones, and residential areas

All these variables are prepared as individual spatial layers covering the entire study area. The relative importance (weight) of each parameter can be assigned using the Fuzzy C-Means (FCM) method to ensure balanced contribution of each factor in the modeling process.

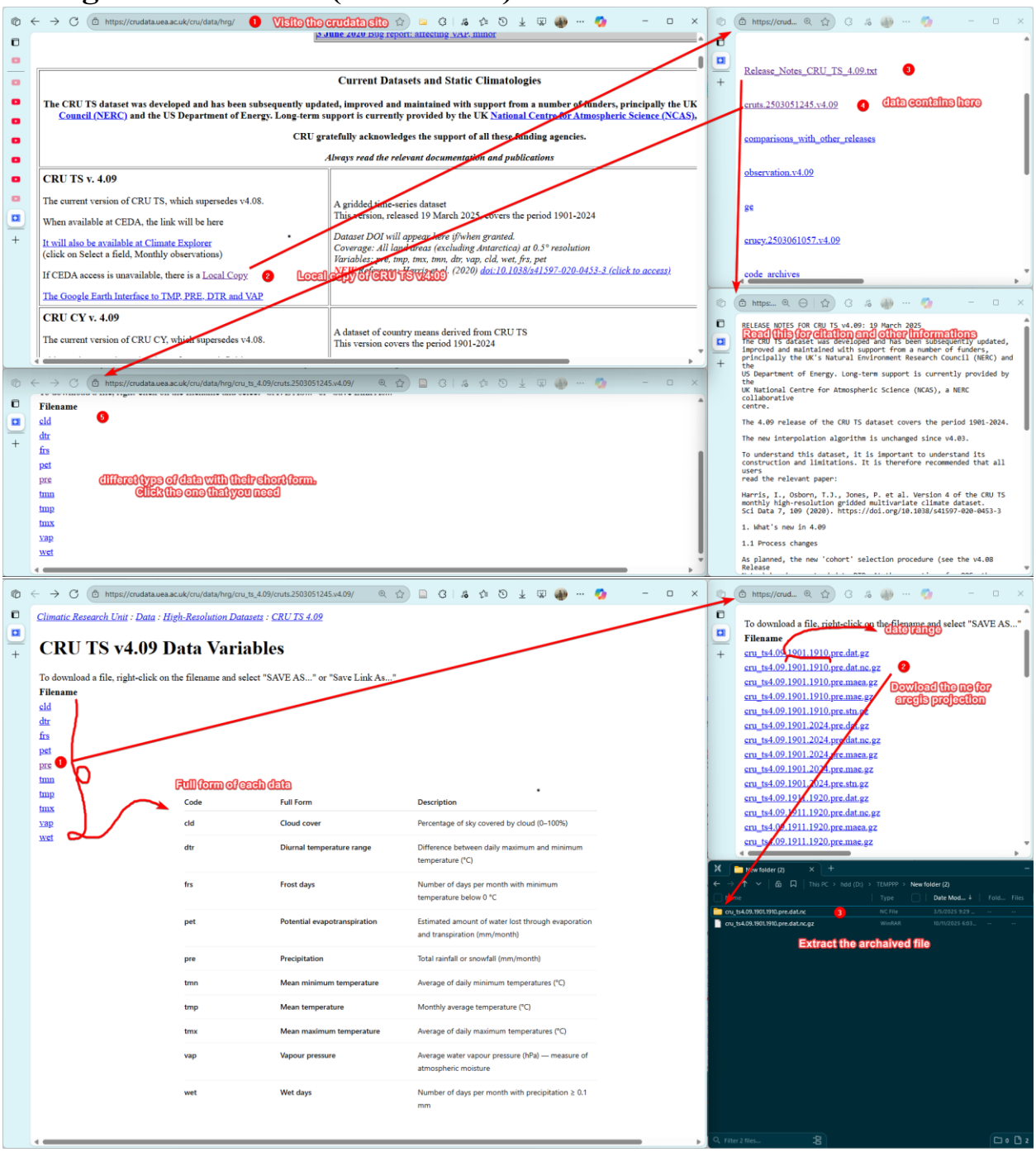
For the species occurrence data, an appropriate sample size should be selected based on the extent of the study area. Typically, 10% of the total occurrence samples are used for smaller regions, while 10–30% of samples are recommended for larger study areas to ensure adequate spatial representation.

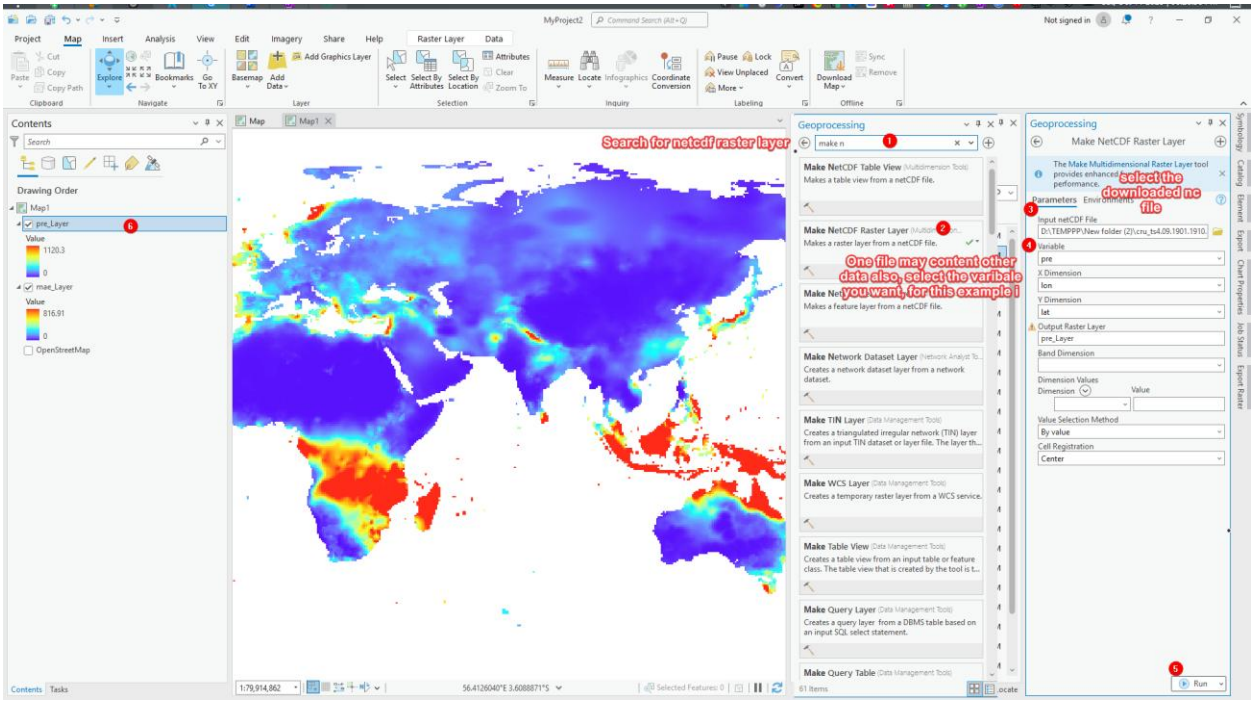


We get 3 types of output from MaxEnt model,

- Contribution table: Shows how much each environmental (or explanatory) variable contributes to the final model prediction.
- Response curve: Shows how the predicted probability of presence (or suitability) changes as a single variable changes, while keeping all others constant.
- Z Chart (Jackknife Test): Assesses the independent and combined influence of each variable on model performance.

How get climate data (Lecture 01)




Search for netcdf raster layer
select the downloaded nc file
One file may content other data also, select the varibale you want, for this example i

Proper way to calculate land cover change, example (Lecture 03)

A Markov transition probability matrix (table) shows the probability of change from one land-cover class to another between two time periods. Below data are assumed for change calculation.

Step 01: Write down the two sets of class areas (pixel)

Class	Legend	Previous_year	Post_year
1	Water	562 701	424 688
2	Tree	393 666	259 607
3	Flooded_veg	269	27
4	Crop	1 352 474	1 302 078
5	Builtup	384 797	711 305
6	Bare	220 049	232 407
7	Range	212 641	196 485

Step 02: Compute overall change ratio for each class.

$ratio = \frac{after}{before}$; for example, for water $\frac{424688}{562701} = 0.755$

Class	Ratio	Interpretation
1	0.755	Water ↓ 24.5 %
2	0.66	Tree ↓ 34 %
3	0.10	Flooded veg ↓ 90 %
4	0.963	Crop ↓ 3.7 %
5	1.848	Built-up ↑ 85 %
6	1.056	Bare ↑ 6 %
7	0.924	Range ↓ 7.6 %

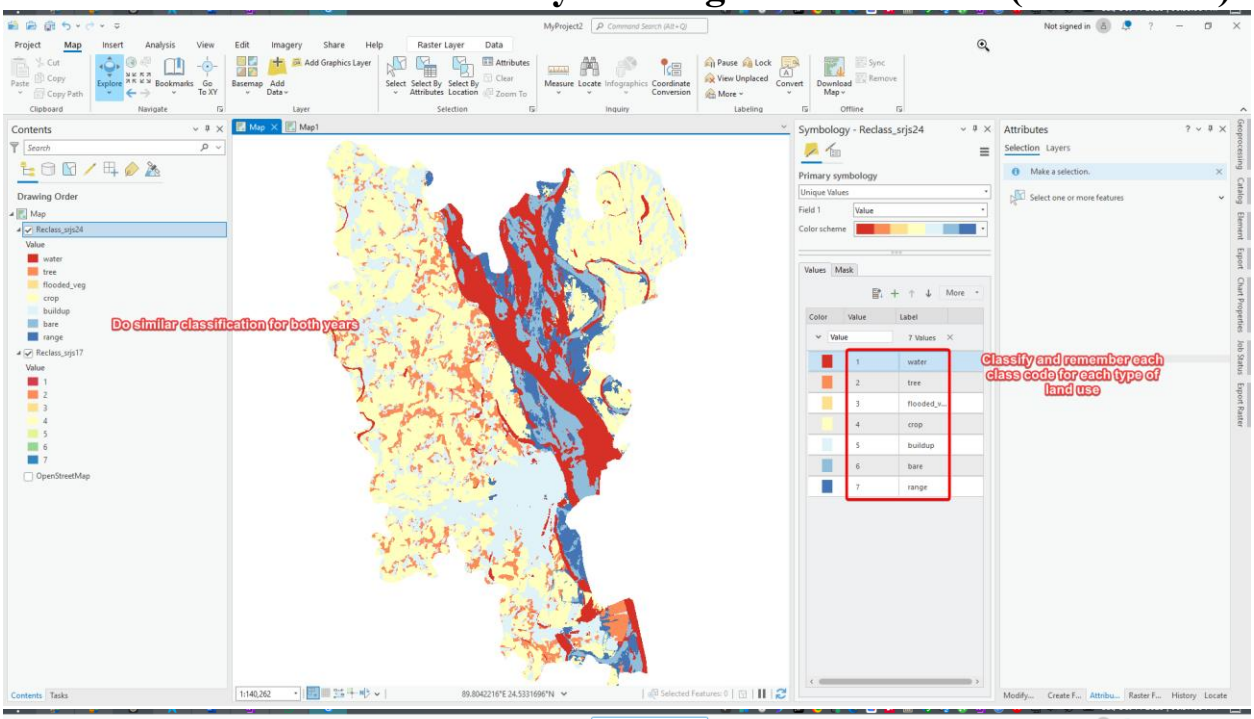
Step 03: Convert to transition probability

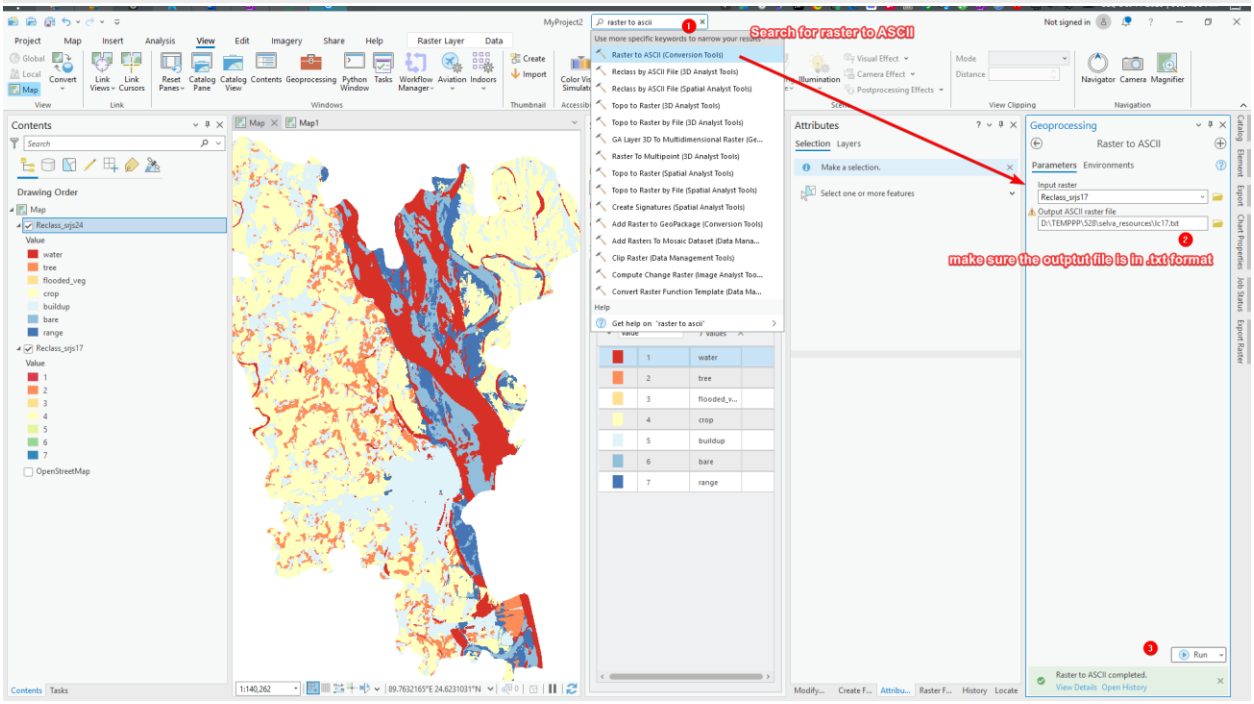
Condition; cell remaining in same class = if ratio<1 and gain portion 1-gain

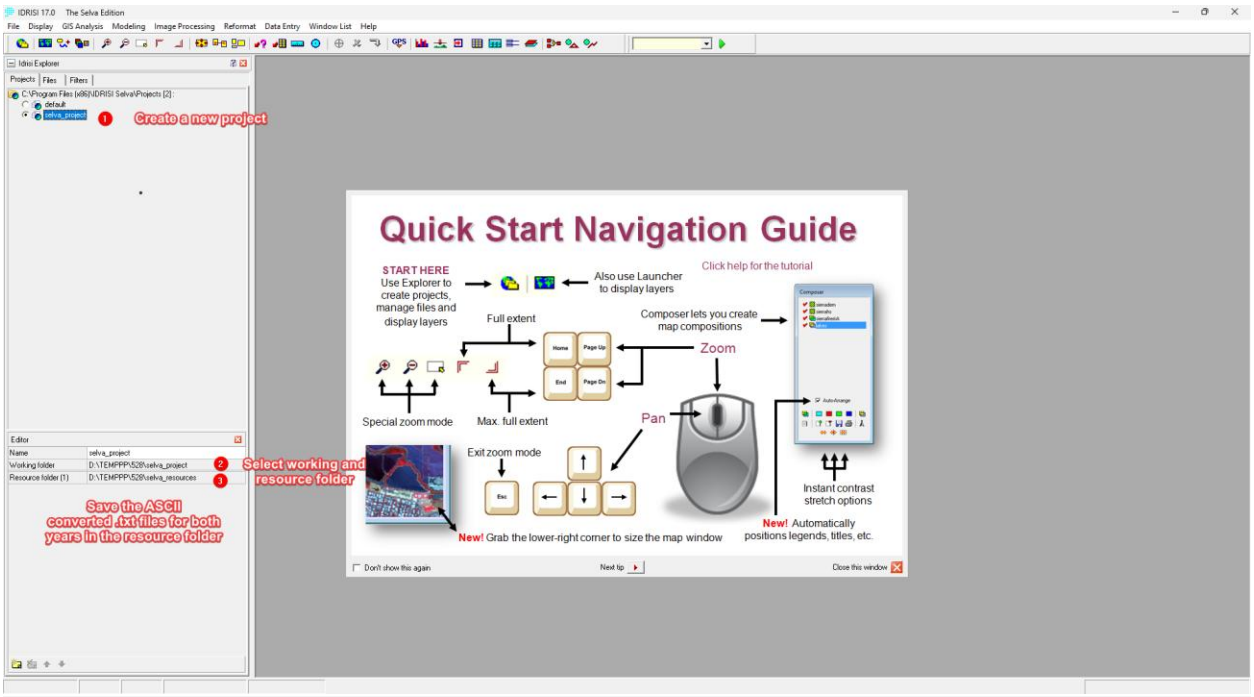
From → To	Water	Tree	Flooded veg	Crop	Built-up	Bare	Range	Total	Loss
Water	**0.755**	0	0	0	*0.197*	*0.049*	0	**1**	**0.245**
Tree	0	**0.66**	0	0	*0.274*	*0.066*	0	**1**	**0.340**
Flooded veg	0.10	0	**0.10**	0	*0.70*	*0.10*	0	**1**	**0.900**
Crop	0	0	0	**0.963**	*0.032*	*0.007*	0	**1**	**0.037**
Built-up	0	0	0	0	**1.000**	0	0	**1**	**0**
Bare	0	0	0	0	0	**1.000**	0	**1**	**0**
Range	0	0	0	0	*0.057*	*0.010*	**0.933**	**1**	**0.076**
Gain	**0.10**	**0**	**0**	**0**	**1.26**	**0.132**	**0**		
Total	**0.955**	**0.66**	**0.1**	**0.963**	**2.26**	**1.132**	**0.933**		

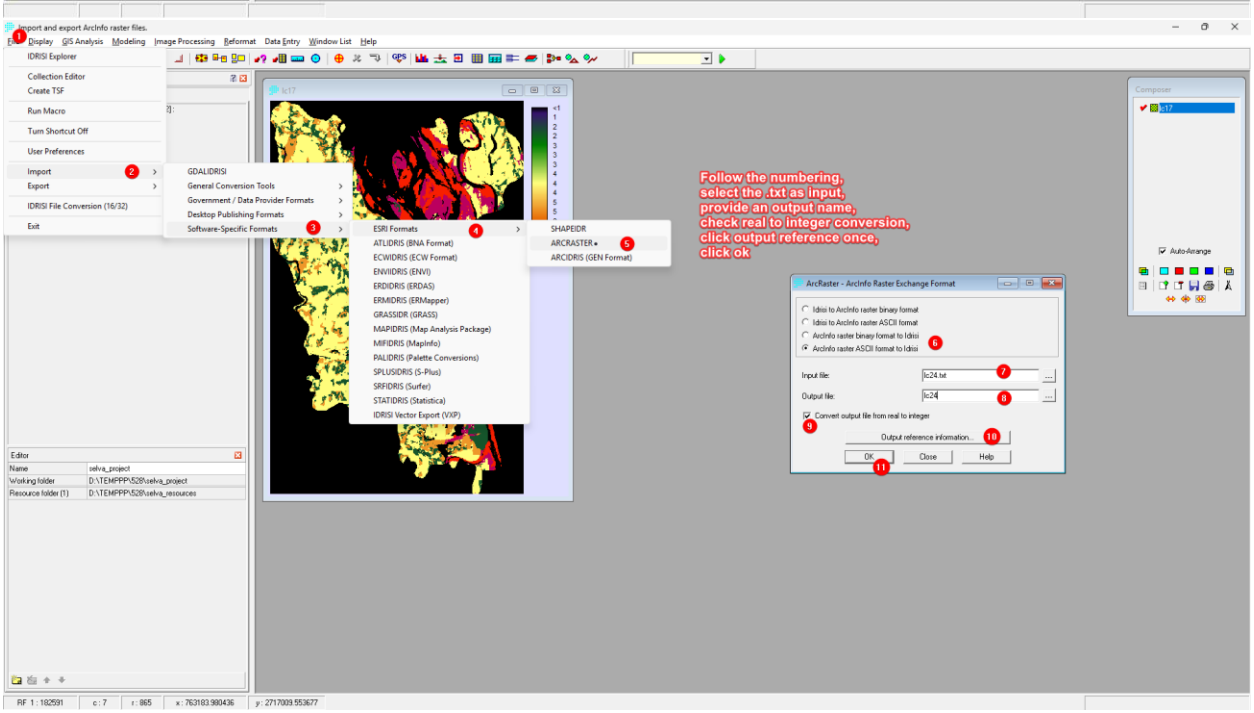
Note: The italicized values are assumed because it's necessary to check how much of one region/pixel is replaced with how much of another region. Rests are calculated based on them.

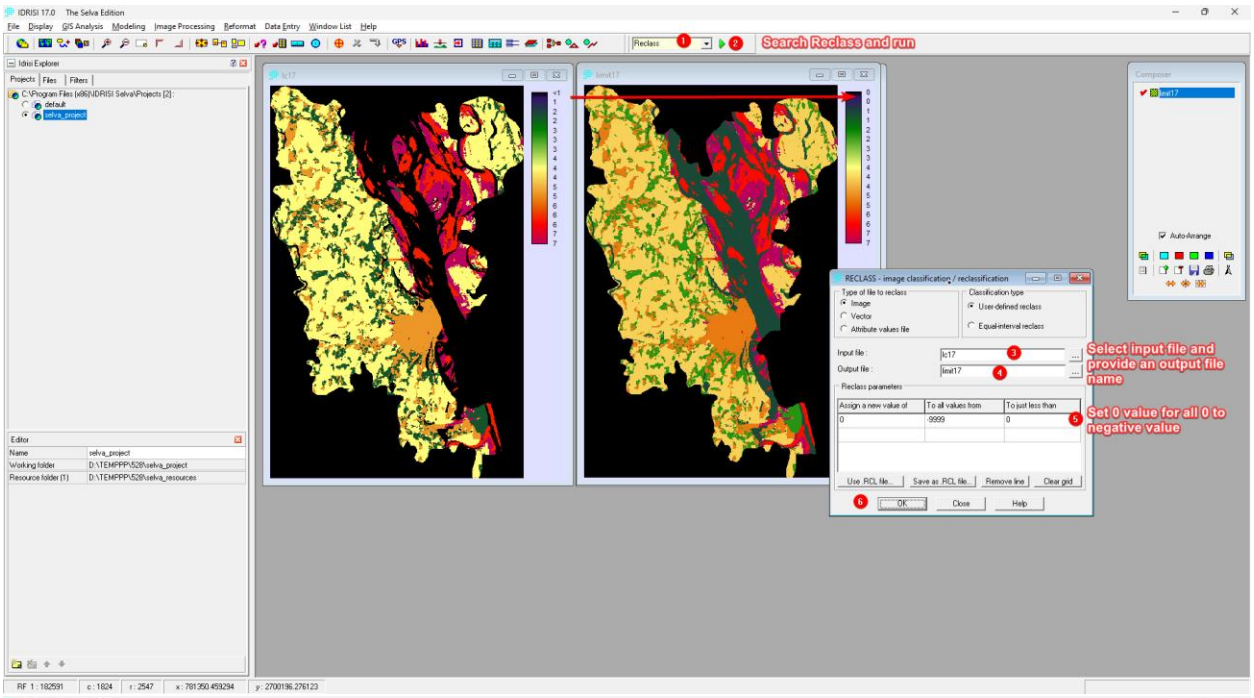
Markov and CA-Markov analysis using Idrisi Selva 17 (Lecture 04)

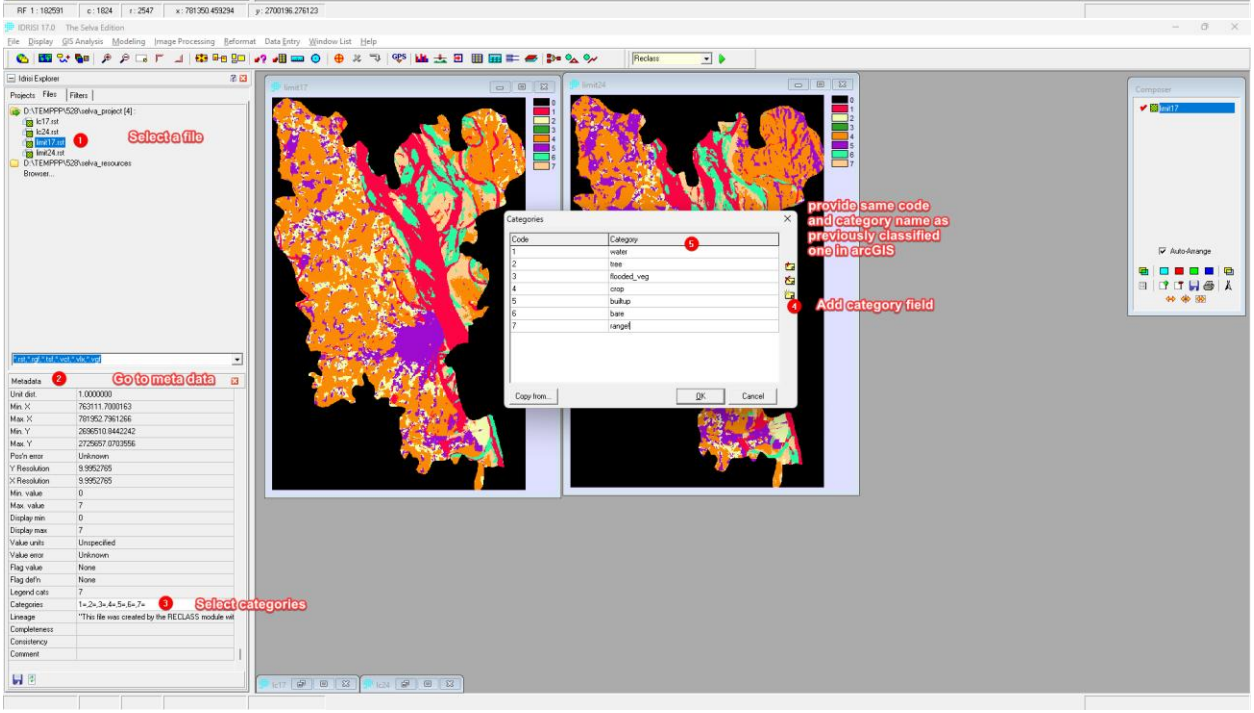


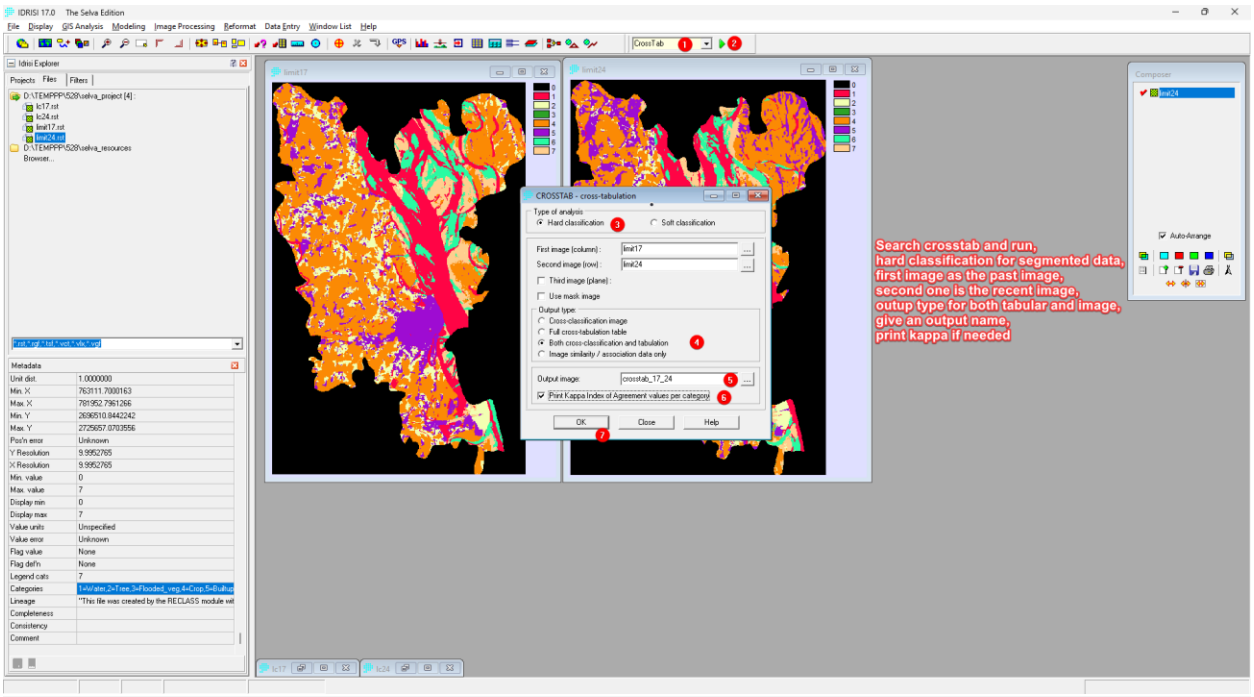


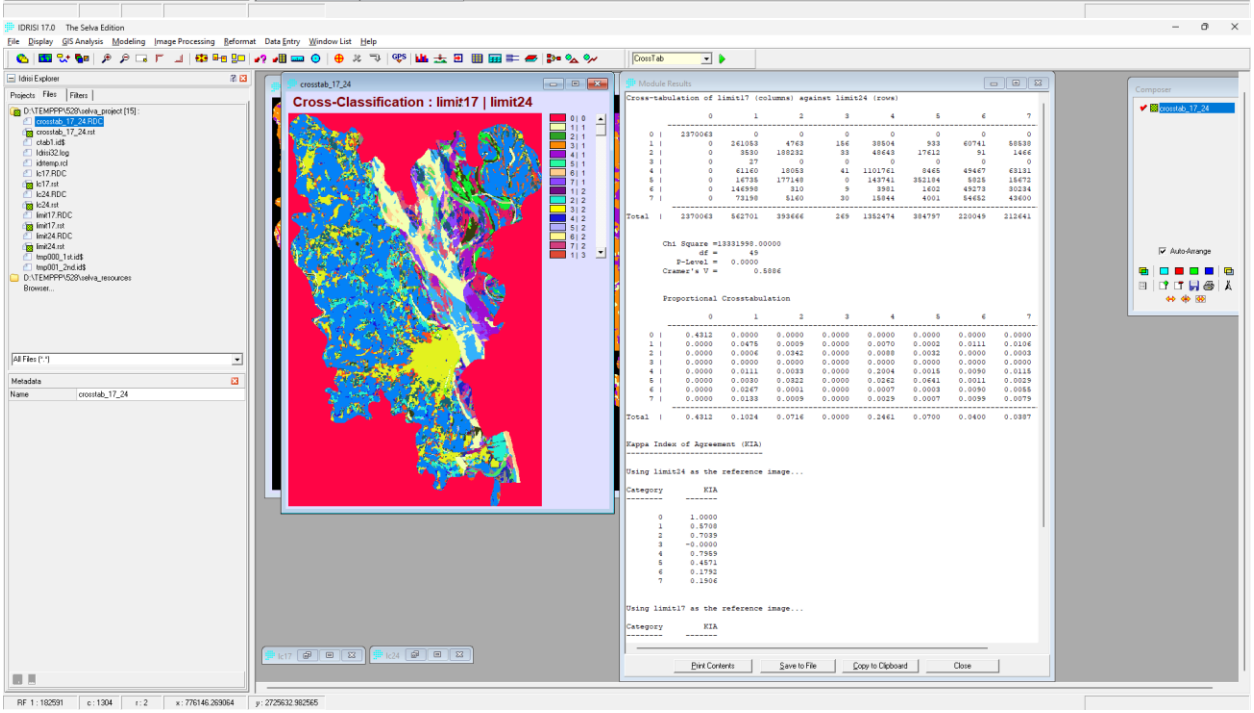

IDRISI 17.0 The Selva Edition
File Display GIS Analysis Modeling Image Processing Reformat Data Entry Window List Help
Idrisi Explorer
Projects Files Filters
C:\Program Files (x86)\IDRISI Selva\Projects [2]:
default
selva_project
1 Create a new project
Quick Start Navigation Guide
START HERE
Use Explorer to create projects, manage files and display layers
Also use Launcher to display layers
Click help for the tutorial
Composer lets you create map compositions
Full extent
Zoom
Special zoom mode
Max. full extent
Pan
Exit zoom mode
Instant contrast stretch options
New! Grab the lower-right corner to size the map window
New! Automatically positions legends, titles, etc.
Don't show this again
Next tip
Close this window
Editor
Name selva_project
Working folder D:\TEMPPP\528\selva_project
Resource folder (1) D:\TEMPPP\528\selva_resources
2 Select working and resource folder
3
Save the ASCII converted .txt files for both years in the resource folder

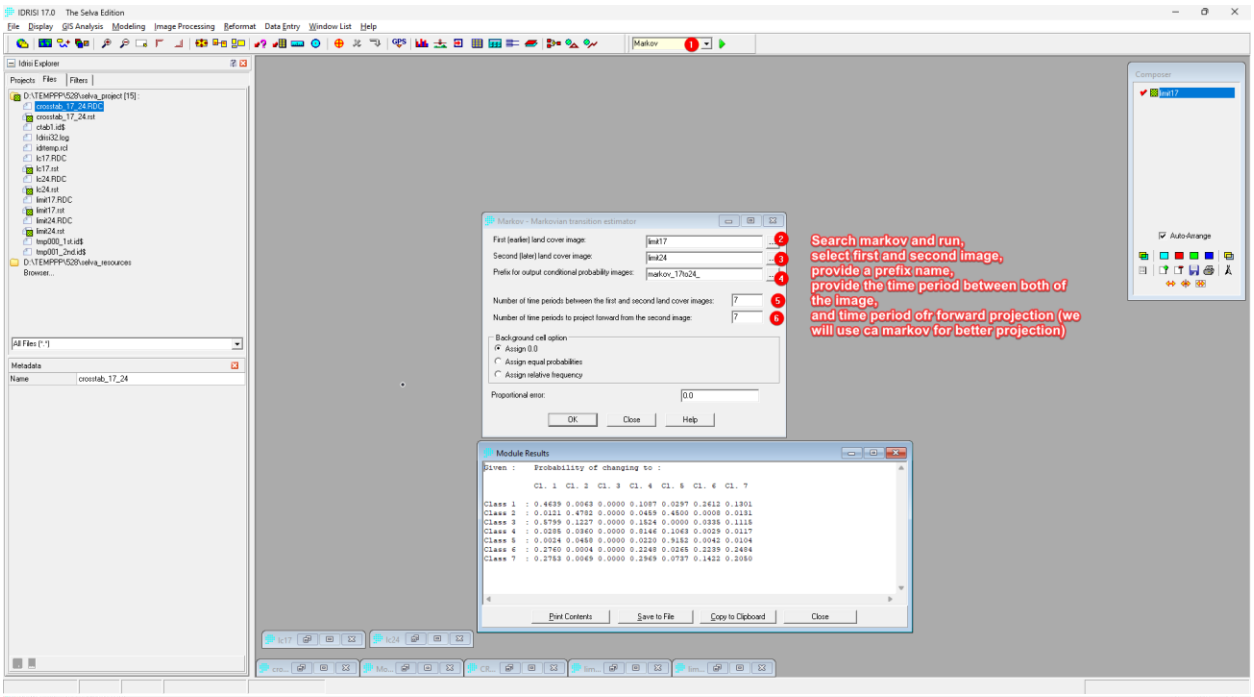

1 Import and export ArcInfo raster files.
Display GIS Analysis Modeling Image Processing Reformat Data Entry Window List Help
IDRISI Explorer
Collection Editor
Create TSF
Run Macro
Turn Shortcut Off
User Preferences
Import 2
Export
IDRISI File Conversion (16/32)
Exit
GDALIDRISI
General Conversion Tools
Government / Data Provider Formats
Desktop Publishing Formats
Software-Specific Formats 3
ESRI Formats 4
ATLIDRIS (BNA Format)
ECWIDRIS (ECW Format)
ENVIIDRIS (ENVI)
ERDIDRIS (ERDAS)
ERMIDRIS (ERMapper)
GRASSIDR (GRASS)
MAPIDRIS (Map Analysis Package)
MIFIDRIS (MapInfo)
PALIDRIS (Palette Conversions)
SPLUSIDRIS (S-Plus)
SRFIDRIS (Surfer)
STATIDRIS (Statistica)
IDRISI Vector Export (VXP)
SHAPEIDR
ARCRASTER 5
ARCIDRIS (GEN Format)
Follow the numbering,
select the .txt as input,
provide an output name,
check real to integer conversion,
click output reference once,
click ok
ArcRaster - ArcInfo Raster Exchange Format
Idrisi to ArcInfo raster binary format
Idrisi to ArcInfo raster ASCII format
ArcInfo raster binary format to Idrisi
ArcInfo raster ASCII format to Idrisi 6
Input file: lc24.txt 7
Output file: lc24 8
Convert output file from real to integer 9
Output reference information... 10
OK 11
Close
Help
Composer
Auto-Arrange
Editor
Name selva_project
Working folder D:\TEMPPP\528\selva_project
Resource folder (1) D:\TEMPPP\528\selva_resources

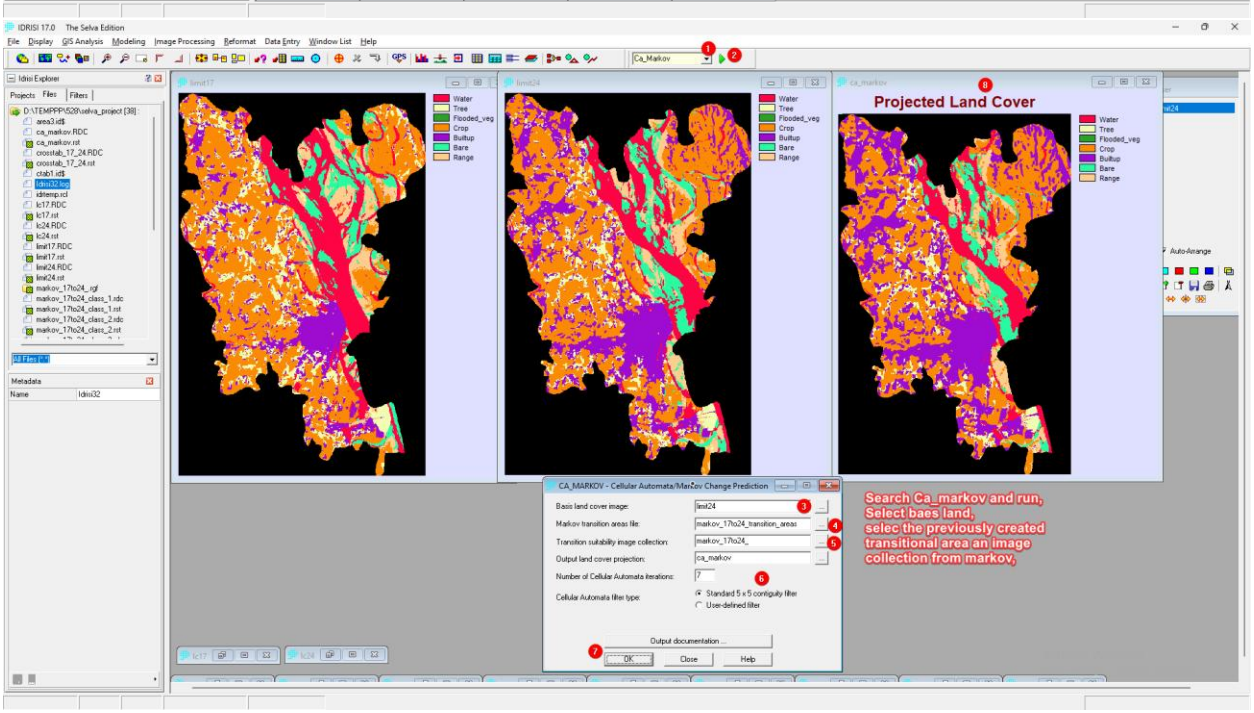

Search Reclass and run
Select input file and provide an output file name
Set 0 value for all 0 to negative value


Select a file
Go to meta data
Select categories
provide same code and category name as previously classified one in arcGIS
Add category field


IDRISI 17.0 The Selva Edition
File Display GIS Analysis Modeling Image Processing Reformat Data Entry Window List Help
CrossTab
Idrisi Explorer
Projects Files Filters
D:\TEMPPP\528\selva_project [4] :
lc17.rst
lc24.rst
limit17.rst
limit24.rst
D:\TEMPPP\528\selva_resources
Browser...
Metadata
Unit dist. 1.0000000
Min. X 763111.7000163
Max. X 781952.7961266
Min. Y 2696510.8442242
Max. Y 2725657.0703556
Pos'n error Unknown
Y Resolution 9.9952765
X Resolution 9.9952765
Min. value 0
Max. value 7
Display min 0
Display max 7
Value units Unspecified
Value error Unknown
Flag value None
Flag def'n None
Legend cats 7
Categories 1=Water,2=Tree,3=Flooded_veg,4=Crop,5=Bulhus
Lineage "This file was created by the RECLASS module wit
Completeness
Consistency
Comment
limit17
limit24
CROSSTAB - cross-tabulation
Type of analysis
Hard classification
Soft classification
First image (column) : limit17
Second image (row) : limit24
Third image (plane) :
Use mask image
Output type:
Cross-classification image
Full cross-tabulation table
Both cross-classification and tabulation
Image similarity / association data only
Output image: crosstab_17_24
Print Kappa Index of Agreement values per category
OK
Close
Help
Search crosstab and run,
hard classification for segmented data,
first image as the past image,
second one is the recent image,
outup type for both tabular and image,
give an output name,
print kappa if needed
Composer
Auto-Arrange


IDRISI 17.0 The Selva Edition
File Display GIS Analysis Modeling Image Processing Reformat Data Entry Window List Help
CrossTab
Idrisi Explorer
Projects Files Filters
D:\TEMPPP\528\selva_project [15] :
crosstab_17_24.RDC
crosstab_17_24.rst
ctab1.idl$
Idrisi32.log
idrtemp.icl
lc17.RDC
lc17.rst
lc24.RDC
lc24.rst
limit17.RDC
limit17.rst
limit24.RDC
limit24.rst
tmp000_1st.idl$
tmp001_2nd.idl$
D:\TEMPPP\528\selva_resources
Browser...
All Files (*.*)
Metadata
Name crosstab_17_24
crosstab_17_24
Cross-Classification : limit17 | limit24
Module Results
Cross-tabulation of limit17 (columns) against limit24 (rows)
Total | 2370063 562701 393666 269 1352474 384797 220049 212641
Chi Square =13331998.00000
df = 49
P-Level = 0.0000
Cramer's V = 0.5886
Proportional Crosstabulation
Total | 0.4312 0.1024 0.0716 0.0000 0.2461 0.0700 0.0400 0.0387
Kappa Index of Agreement (KIA)
Using limit24 as the reference image...
Category KIA
0 1.0000
1 0.5708
2 0.7039
3 -0.0000
4 0.7959
5 0.4571
6 0.1792
7 0.1906
Using limit17 as the reference image...
Category KIA
Print Contents
Save to File
Copy to Clipboard
Close
Composer
Auto-Arrange
RF 1 : 182591 c : 1304 r : 2 x : 776146.269064 y : 2725632.982565


IDRISI 17.0 The Selva Edition
Markov
Markov - Markovian transition estimator
First (earlier) land cover image: limit17
Second (later) land cover image: limit24
Prefix for output conditional probability images: markov_17to24_
Number of time periods between the first and second land cover images: 7
Number of time periods to project forward from the second image: 7
Background cell option
Assign 0.0
Assign equal probabilities
Assign relative frequency
Proportional error: 0.0
OK
Close
Help
Module Results
Given : Probability of changing to :
Print Contents
Save to File
Copy to Clipboard
Close
Search markov and run,
select first and second image,
provide a prefix name,
provide the time period between both of the image,
and time period ofr forward projection (we will use ca markov for better projection)


IDRISI 17.0 The Selva Edition
Ca_Markov
Projected Land Cover
Water
Tree
Flooded_veg
Crop
Builtup
Bare
Range
CA_MARKOV - Cellular Automata/Markov Change Prediction
Basis land cover image: limit24
Markov transition areas file: markov_17to24_transition_areas
Transition suitability image collection: markov_17to24_
Output land cover projection: ca_markov
Number of Cellular Automata iterations: 7
Cellular Automata filter type:
Standard 5 x 5 contiguity filter
User-defined filter
Output documentation ...
OK
Close
Help
Search Ca_markov and run,
Select baes land,
selec the previously created transitional area an image collection from markov,